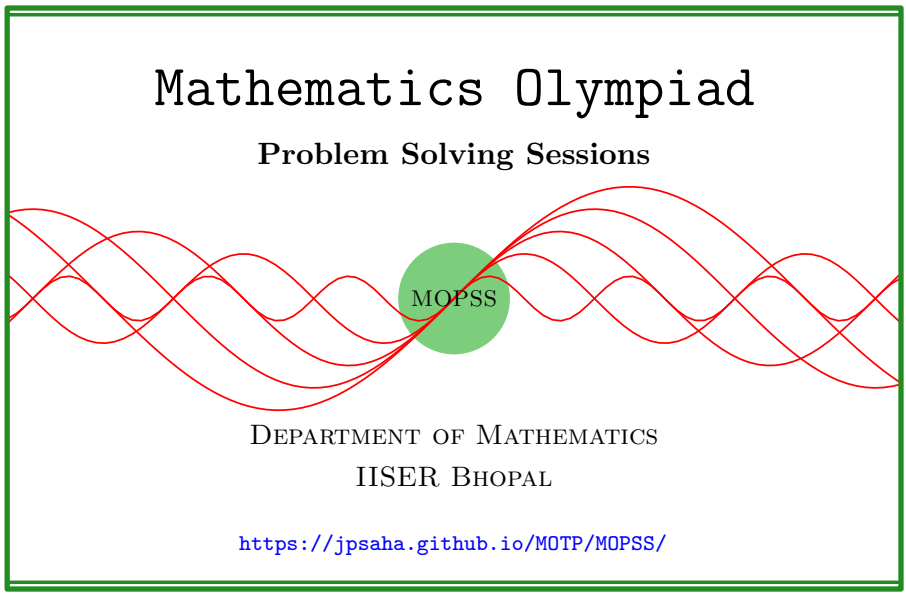
Warm up

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (All-Russian Mathematical Olympiad 1999 Grade 9 Day 1 P1, , by S. Volchenkov) 2
2 Exercise (Bay Area MO 2004 P4,) 2

§1 Warm up

Exercise 1 (All-Russian Mathematical Olympiad 1999 Grade 9 Day 1 P1, , by S. Volchenkov)**.** The decimal digits of a natural number A form an increasing sequence (from left to right). Find the sum of the digits of $9A$.

> **Walkthrough** —
> **(a)**

Solution 1.

■

Exercise 2 (Bay Area MO 2004 P4,)**.** Suppose one is given n real numbers, not all zero, such that their sum is zero. Prove that one can label these numbers $a_1, a_2, \ldots, a_n$ in such a manner that

$$a_1a_2 + a_2a_3 + \cdots + a_{n-1}a_n + a_na_1 < 0.$$

> **Walkthrough** — Consider the sums for all possible labelings of the numbers.

Solution 2. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)